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DEPARTMENT OF ENGINEERING SCIENCES

A

PROJECT REPORT

Under Project Based Learning and Management Course (PBLM-II) on

SMART TRACKING & ALERT for LEOPARD KINETICS (S.T.A.L.K.)

FY B - Tech IT, Division 'C'

SUBMITTED BY

Sr. No	Name of Student	PRN No
01	Mr. MAYUR KAPSE	24510060
02	Mr. HEMANSHU KAWDO	24510043
03	Mr. SWAROOP KARNEWAR	24510061
04	Mr. SHREYASH KAMTHE	24510059
05	Mr. SIDDESH KHIRATKAR	24510121
06	Mr. ATHARVA JAWADE	24510008

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DEPARTMENT OF ENGINEERING SCIENCES

Institute Vision

To be recognized amongst top 10 private engineering colleges in Maharashtra by the year 2026 by rendering value added education through academic excellence, research, entrepreneurial attitude and global exposure.

Institute Mission

1. To Enable Placement Of 150 Plus Students in The 7 Lacs Plus Category & Ensure 100% Placement of All Eligible Final Year Students.

2. To Connect With 10 Plus International Universities, Professional Bodies and Organizations To Provide Global Exposure To Students.

3. To Create Conducive Environment For Career Growth, Prosperity, And Happiness Of 100% Staff.

4. To Be Amongst Top 5 Private Colleges In Pune In Terms Of Admission Cut Off.

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Program Outcomes (POs)

PO1	**Engineering knowledge**	Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	**Problem analysis**	Identify, formulate, research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences
PO3	**Design/ development of solutions**	Design solutions for complex engineering problems and design system components, processes to meet the specifications with consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	**Conduct investigations of complex problems**	Use research-based knowledge including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	**Modern tool usage**	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	**The engineer and society:**	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	**Environment and sustainability**	Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	**Ethics**	Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	**Individual and team work**	Function effectively as an individual, and as a member or leader in teams, and in multidisciplinary settings.

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PO10	**Communication**	Communicate effectively with the engineering community and with society at large. Be able to comprehend and write effective reports documentation. Make effective presentations, and give and receive clear instructions.
PO11	**Project management and finance**	Demonstrate knowledge and understanding of engineering and management principles and apply these to one's own work, as a member and leader in a team. Manage projects in multidisciplinary environments.
PO12	**Life-long learning**	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

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DEPARTMENT OF ENGINEERING SCIENCE

Academic Year 2024-25

CERTIFICATE

This is to certify that,

Project Report entitled, "**SMART TRACKING & ALERT for LEOPARD KINETICS (S.T.A.L.K.)**"

Submitted by

Sr.no.	Name of Students	PRN Number	Sign
1	**Mr. MAYUR KAPSE**	24510060	
2	**Mr. HEMANSHU KAWDO**	24510043	
3	**Mr. SWAROOP KARNEWAR**	24510061	
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5	**Mr. SIDDESH KHIRATKAR**	24510121	
6	**Mr. ATHARVA JAWADE**	24510008	

has satisfactorily completed the **term work/practical/Oral** in the Course of **Problem Based Learning Management II (FEHSM209)** as per the Autonomy Curriculum of AISSMS's Institute of Information Technology, Pune during the **Academic Year, 2024-25 Semester II**.

Date:

Prof N.D. Gaikwad

Guide

Dr. P. G. Mushrif

Head of Department

External Name and Sign

ACKNOWLEDGEMENT

It is my great pleasure in expressing sincere and deep gratitude towards my guide Mr. N. D. Gaikwad , Assistant Professor Engineering Sciences Department for her/his valuable guidance and constant support throughout this work and help
to peruse additional studies "SMART TRACKING & ALERT for LEOPARD KINETICS (S.T.A.L.K.) ".

We take this opportunity to thank Head of the Department Dr. Pramod Musrif and COURSE coordinator Dr. Y.P. Patil and all Mentor staff members of department of Engineering Sciences, AISSMS IOIT, Pune, for cooperation provided by them in many ways. The motivation factor for this work was the inspiration given to us by our Honourable Principal Dr. P. B. Mane.

I would like to thank our Guide Mr. N. D. Gaikwad and mentors for the project idea and for providing data. We would like to thank for her/his immense help and numerous suggestions during my work and verify result. Lastly ,I am thankful to those who have directly or indirectly supported for our work.

STUDENT UNDERTAKING

We, the undersigned Students of FY BTech, hereby declare that the project titled "SMART TRACKING & ALERT for LEOPARD KINETICS (S.T.A.L.K.)" is our original work, conducted under the guidance of Prof. N. D. Gaikwad. This project has been undertaken in partial fulfilment of the requirements for Problem Based Learning Management II (FEHSM209) as per the Autonomy Curriculum of AISSMS's Institute of Information Technology, Pune during the Academic Year 2024-25 Semester II.All sources of information and data used have been appropriately cited and acknowledged.

We further declare that this project has not been submitted elsewhere, in whole or in part, for the award of any other degree or diploma. We also declare that we have adhered to all academic, engineering professional, and integrity standards and have not fabricated or falsified any ideas, data, or sources in our submission. We are aware that any violation of ethical standards may cause disciplinary action as per University/Institute rules and regulations.

Date: ___________________________________

Place: __________________________________

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ABSTRACT

This real-time leopard detection system presents a proactive approach to managing human-wildlife conflicts, especially in urban areas bordering forested regions. The project integrates artificial intelligence and computer vision to detect leopards near human settlements and send instant SMS alerts with location details, helping to prevent potential harm to both humans and wildlife.

At the core of the system is the YOLOv5 object detection model, trained specifically on leopard images for high accuracy and speed. The dataset was prepared and labelled using Roboflow, ensuring the model was optimized for robust detection performance. Training was conducted on Google Colab, and the best-performing model weights were used for real-time inference.

The final system operates using a standard webcam feed. When a leopard is detected in the video stream, an SMS alert with the geographic location is automatically sent to relevant authorities or stakeholders. This alert-based mechanism ensures immediate action without requiring physical alarms or on-site personnel.

Designed to be lightweight, scalable, and field-deployable, this system demonstrates how AI-powered monitoring can be effectively applied in wildlife protection. It not only safeguards human lives but also contributes to conservation efforts through early detection. The approach is also adaptable for monitoring other species in various ecological or safety contexts.

TABLE OF CONTENTS

INTRODUCTION

With the growing need for wildlife conservation and the rise in human-wildlife conflicts, technology-driven approaches have gained importance—particularly those using artificial intelligence and computer vision. Object detection models like YOLO (You Only Look Once) have become central to real-time animal monitoring due to their high speed and accuracy.

YOLOv5, a lightweight and fast version, offers easy deployment and real-time detection, making it suitable for field applications. This project builds on such advancements, using YOLOv5 for leopard detection in video feeds, combined with an SMS alert system for quick response.

A critical part of the project involved preparing a clean and well-labeled dataset. Tools like Roboflow enabled dataset augmentation, annotation, and format conversion, which are essential for high-performance training. Literature also points to the success of AI-based alert systems in detecting animals and notifying people in real time, thus reducing incidents of conflict.

This review brings together key research on object detection, dataset management, and real-time alert systems, forming the foundation of our project's technical approach and implementation strategy.

LITERATURE RIVEWED

Sr. No.	Name / Author	Paper Name	Parameter / Technology Studied	Conclusion / Remark	Future Scope
1	Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016)	*You Only Look Once: Unified, Real-Time Object Detection*	Introduction of the YOLO algorithm for real-time object detection	YOLO enables high-speed detection with good accuracy	Enhancements for small object detection and real-time field deployment
2	Jocher, G. et al. (2020)	*YOLOv5 by Ultralytics (GitHub)*	Lightweight and fast object detection architecture	YOLOv5 improves speed, simplicity, and training efficiency	Expanding to various real-time detection use-cases like wildlife alerts
3	Smith, L., Gupta, A., & Rao, M. (2022)	*Efficient Dataset Annotation using Roboflow for Animal Detection*	Dataset labeling, augmentation & export for YOLO	Roboflow greatly simplifies dataset prep and annotation	Potential to automate dataset creation across species and geographies
4	Miller, D. A., Patel, J., & Zhang, Y. (2019)	*AI in Conservation: Deep Learning Applications in Wildlife Monitoring*	Use of AI with camera traps and object detection	AI can help monitor endangered species effectively	Can support real-time alert systems in conflict-prone zones
5	Singh, R., Iyer, K., & Verma, P. (2022)	*Mitigating Human-Wildlife Conflicts with AI-Based Detection Systems*	AI-based detection to reduce conflict & alert communities	Shortens response time to wildlife intrusion	Useful for scalable, multi-animal detection in rural and semi-urban areas

SUMMARY

Real-Time Leopard Detection Using YOLO and IoT-Based Alert System

In response to the growing threat of leopard intrusions in semi-urban areas, a project by Arnav Deep and Manjot Singh from IIITDM Jabalpur showcases a practical implementation of a real-time object detection and alert system designed to identify and respond to leopard sightings. The need for such a system arose from frequent leopard incursions on their institute's campus, located near the Dumna Nature Reserve, a known wildlife corridor.

The team employed a custom-trained object detection model using Darkflow, a simplified TensorFlow implementation of the YOLO (You Only Look Once) framework. Their model was specifically trained to recognize leopards using a dataset compiled from open sources and wildlife imagery. The focus was on building a system that is lightweight and efficient enough to operate on embedded hardware.

The detection system was deployed on a Raspberry Pi platform using a Pi NoIR (infrared) camera, allowing for day and night surveillance. A GSM module was integrated to send SMS alerts to residents and security personnel when a leopard was detected. The combination of computer vision, machine learning, and embedded systems highlights the potential of low-cost hardware in addressing serious wildlife safety issues.

What makes this project significant is its real-world applicability. It demonstrates that affordable, real-time detection systems can be built with open-source tools and low-power devices, which is especially useful in wildlife-prone or rural areas lacking high-end surveillance infrastructure.

Citation:

Arnav Deep & Manjot Singh. *Leopard Detection Using YOLO on Embedded System*. GitHub Repository. Retrieved from https://github.com/arnav-deep/leopard-detection

PROBLEM STATEMENT

The increasing frequency of leopard attacks in India has raised serious concerns regarding human-wildlife conflicts, especially in rural and semi-urban areas that lie in proximity to wildlife sanctuaries and forested regions. Over the past several years, incidents of leopards venturing into human settlements have been reported in various states, including Maharashtra, Madhya Pradesh, Uttar Pradesh, and others. These attacks are a direct result of the continuous encroachment of human activities into the natural habitats of leopards, which forces these animals to venture into populated areas in search of food and territory.

Leopards, although generally shy and solitary animals, are opportunistic hunters, and their natural prey is increasingly becoming scarce due to deforestation, urbanization, and other human activities that destroy their habitats. Consequently, the leopards are often compelled to enter residential areas, where they come into direct contact with humans, leading to potentially fatal attacks. These conflicts are not only dangerous to humans but also result in the harm or death of the animals themselves, as they are sometimes forced to attack livestock or are killed in retaliation by the local population.

The attacks typically occur when the leopards come out at night or early morning, when it is dark and human activity is minimal. These attacks have led to tragic fatalities, especially among children, as well as serious injuries in many cases. Victims of leopard attacks often suffer from extensive mauling, and the psychological trauma of these incidents affects entire communities. Additionally, there is growing frustration among residents of villages and towns near forests, who feel increasingly vulnerable to these attacks.

Leopard Attack Incidents in India (2023-2025)

INCIDENTS REPORTS

1. Jannar, Maharashtra (December, 2024)

In December 2024, a devastating attack in Jannar division, Maharashtra, resulted in the death of a 4-year-old girl. This was part of a broader trend in the region, where several leopard attacks were reported in the same year, leading to nine fatalities. The increasing frequency of such incidents has been attributed to the loss of the leopard's natural habitat, combined with the encroachment of human settlements into forested areas. The leopards, in search of food, have been forced to leave the forest and enter populated spaces.
Source *- Indian Express*

2. Sheopur, Madhya Pradesh (March, 2025)

In March 2025, a 9-year-old boy in Sheopur, Madhya Pradesh, was attacked by a leopard near Kuno National Park. The child was dragged away by the leopard, but his mother heroically fought the animal off, saving his life. However, the boy sustained over 120 injuries and underwent several surgeries. This incident highlights the increasing risk of such attacks in areas where leopards are pushed into closer contact with human settlements due to decreasing prey populations in their natural habitats.

Source - Times of India

3. Bahraich, Uttar Pradesh (January to November, 2024)

Between January and November 2024, Bahraich district in Uttar Pradesh witnessed a disturbing surge in leopard attacks, resulting in multiple fatalities. Victims ranged from young children to adults, with a 32-year-old man also falling victim. These attacks have been attributed to a combination of deforestation, encroachment of human settlements into the leopards' natural territories, and the diminishing prey available for leopards in forested areas. The local population, living in close proximity to the forest, is increasingly at risk.

Source - Times of India

4. Ratnagiri, Maharashtra (April, 2023)

A heartbreaking incident occurred in April 2023 in Ratnagiri, Maharashtra, when a 10-year-old girl was killed in a leopard attack. The child was outside her home when the leopard attacked, dragging her into the forest. This tragic event is part of a growing trend in the region, where leopards are encroaching on human settlements due to deforestation and a lack of prey. The loss of human lives has sparked local protests and demands for stronger measures to mitigate leopard attacks and ensure the safety of vulnerable communities.

OBJECTIVES

1. Real-Time Leopard Detection and Early Warning System
   - The system will utilize YOLOv5 for detecting leopards in images and video feeds in real-time. Upon detecting a leopard, an early warning system will notify local authorities and nearby communities, preventing potential conflicts.

2. Data Collection and Training
   - A large dataset of leopard images and videos will be compiled and labelled using Roboflow. This data will be used to train the detection model, improving its accuracy and enabling it to recognize leopards in diverse environments.

3. User Accessibility and System Integration
   - A simple, intuitive web-based interface will be developed for easy interaction. The system will also integrate real-time alert mechanisms, such as mobile apps or text notifications, to ensure immediate response when a leopard is detected.

4. Scalability and Continuous Improvement
   - The system will be designed to be scalable for deployment in different regions facing similar wildlife conflicts. Additionally, continuous improvements will be made based on user feedback and new data to enhance detection accuracy.

5. Community Engagement and Feedback
   - The project will actively involve local communities to gather feedback and ensure the system meets their needs.

PROJECT DEVELOPMENT FLOWCHART

Identified the Problem Statement

↓

Formed a Team

↓

Discussion on Project

↓

Divided the Work & Assigned roles

↓

Searched for Resources

↓

Image Labelling with Roboflow

↓

Model Training on Roboflow & Google Colab

↓

Testing and Error Correction

↓

Integration of Detection System

↓

Automated Alert System Implementation

↓

Final Testing

SOFTWARE STACK AND CODE INTEGRATION

A] Dataset Curation, Preparation and Exportation Using Roboflow

1. Dataset Curation in Roboflow

We used Roboflow to organize and prepare our dataset for leopard detection:

a. Custom Toy Leopard Images

- Manually captured multiple images of a toy leopard from various angles and lighting conditions.
- Annotated each image using bounding boxes in Roboflow.
- These helped in demonstrating object detection during system testing.

b. Real Leopard Dataset from Roboflow Universe

- Searched for a publicly available real leopard dataset.
- Selected a diverse dataset containing multiple backgrounds and postures.
- This ensured our model learned from real-world conditions.

2. Dataset Merging and Annotation

- Combined toy and real leopard datasets into a unified Roboflow project.
- Standardized all labels to leopard for training consistency.
- Verified each bounding box manually to ensure precise annotations.
- Accurate labeling is critical for YOLOv5, which uses .txt files for bounding boxes and class pairing.

3. Preprocessing and Augmentation

Used Roboflow's tools to improve model generalization:

- Image resizing to 416×416 and 640×640 (YOLOv5-compatible).
- Augmentations applied:
  - Rotation (±15°)

  - Flipping (horizontal/vertical)
  - Brightness/contrast changes
  - Noise addition

4. Dataset Export for YOLOv5

- Exported the dataset in YOLOv5 PyTorch format, which included:
  - train/, valid/, test/ folders with images and .txt labels.
  - A data.yaml file defining class and folder paths.
- Optimized image size and file count for faster model loading and effective training.

5. API Key and Integration

- Generated a private Roboflow API key.
- Enabled automatic dataset import into Google Colab via a Python script.
- Eliminated the need for manual upload and always accessed the latest dataset version.

```
from roboflow import Roboflow
rf = Roboflow(api_key="YOUR_API_KEY")
project = rf.workspace("your-workspace").project("leopard-detection")
dataset = project.version(x).download("yolov5")
```

B] Model Training and Evaluation Using Google Colab

1. Environment Setup

- We used Google Colab for training due to free GPU access and robust Python support.
- Cloned the YOLOv5 GitHub repository:

```
!git clone https://github.com/ultralytics/yolov5
%cd yolov5
```

- Installed all required packages and dependencies including from requirements.txt:

```
!pip install roboflow
!pip install -r requirements.txt
```

2. Dataset Integration via Roboflow API

- Used our Roboflow API key to pull the dataset directly into Colab after installing roboflow:

```
!pip install roboflow
from roboflow import Roboflow
rf = Roboflow(api_key="your_api_key")
project = rf.workspace().project("leopard-detection")
dataset = project.version("1").download("yolov5")
```

- The download included:
  - train/, valid/, test/ folders
  - data.yaml containing dataset metadata and class info

3. Model Training

We trained using the yolov5s (small) model, which balances speed and accuracy:

```
!python train.py --img 640 --batch 16 --epochs 50 --data dataset/data.yaml --weights yolov5s.pt --name leopard_model
```

Explanation of Arguments:

- --img 640: Sets input image resolution to 640×640.
- --batch 16: Number of images processed at once.
- --epochs 50: Model trains for 50 complete passes over the dataset.
- --data: Points to the data.yaml file.
- --weights yolov5s.pt: Starts training from pretrained YOLOv5s weights (on COCO).
- --name leopard_model: Saves results in a folder named leopard_model.

4. Monitoring Training

YOLOv5 provides:

- Real-time graphs for loss (box, object, classification).
- mAP, precision, and recall scores.
- A final results.png chart summarizing performance.

5. Saving and Exporting Weights

- Best-performing weights saved automatically as best.pt:

```
runs/train/leopard_model/weights/best.pt
```

- Transferred the file to Google Drive for easy backup and future use:

C] SMS Alert Integration Using Twilio

1. Purpose

To immediately notify relevant authorities or individuals when a leopard is detected.

2. Technology Used: Twilio

- Twilio was used for sending SMS alerts.
- Required Twilio Account SID, Auth Token, and Twilio phone number.
- Mainly due to less price for sim card number and its trials.

3. SMS Format

The alert message sent to recipients followed this format:

```
🚨 Leopard Detected!

Location: https://www.google.com/maps?q=18.531669,73.867227

Please respond immediately.
```

(This message was used in real testing and reflects what recipients would receive.)

4. Python Code for Alert:

```
from twilio.rest import Client

account_sid = 'your_account_sid'
auth_token = 'your_auth_token'
client = Client(account_sid, auth_token)

def send_sms_alert(to_number):
    message_body = (
        "🚨 Leopard Detected!\n\n"
        "Location: https://www.google.com/maps?q=18.531669,73.867227\n\n"
        "Please respond immediately."
    )
    client.messages.create(
        body=message_body,
        from_='+1234567890',
        to=to_number
    )

if leopard_detected:
    recipient_number = '+919876543210'
    send_sms_alert(recipient_number)
```

FUTURE APPLICATION AND SCOPE

Our leopard detection system can be used in wildlife conservation efforts to monitor animal movement in forests and protected areas. By using webcams and AI models like YOLOv5, authorities can receive instant SMS alerts when a leopard is detected. This helps prevent poaching and keeps track of endangered animals more effectively.

It can also help reduce human-wildlife conflicts. In many rural areas, leopards enter villages or farms. A system like ours can warn people early by sending alerts, giving them time to stay safe or take action. Similar AI-based systems have already helped reduce leopard attacks in some regions of Maharashtra.

In farming, our model can be retrained to detect livestock and protect them from wild predators like jackals or leopards. Farmers can be alerted immediately when such animals are nearby.

Urban areas and zoos can also benefit from this technology. Sometimes wild animals enter parks or residential areas, causing panic. Our system can detect them early and notify the authorities. It can also be used inside zoos to monitor animal enclosures.

Finally, researchers can use the system to study animal behavior. By adding GPS and timestamps, the system can help track movement patterns or migration..

CHALLENGES FACED DURING IMPLEMENTATION

While the project was successfully completed, we faced several challenges across data collection, model training, real-time detection, and alert integration.

One major challenge was working with the dataset. Although Roboflow Universe offered leopard datasets, many had inconsistent or unclear labels like "big_cat" or "panthera," which caused issues during model training. We had to standardize all labels to "leopard" to ensure proper learning. Some images were poorly annotated or had incorrect bounding boxes, especially when the leopard was camouflaged. This required manual correction of many annotations, including for toy leopard images used during early testing.

Training the model on Google Colab also came with its own set of difficulties. Free GPU sessions were time-limited, and larger training jobs often got interrupted. We had to train in smaller batches or resume from saved checkpoints. Downloading the trained model (best.pt) was sometimes slow or failed due to session disconnections, so we used Google Drive for smoother file handling.

During real-time detection, integrating the webcam with our script had some hiccups. Initially, the camera wasn't detected properly, and we had to debug driver and access permissions. Once it worked, video feed lag became a problem due to Python processing overhead, which slightly affected detection speed and accuracy.

Lastly, setting up SMS alerts via Twilio required precise formatting and testing. We had to configure the message content, GPS location, and recipient details to ensure the alerts were timely and readable

CONCLUSION

This project set out to create an affordable, AI-powered wildlife monitoring system capable of detecting leopards in real time. By combining tools like Roboflow for dataset preparation, YOLOv5 for object detection, and Google Colab for model training, we successfully developed a working prototype that demonstrates the potential of machine learning in wildlife surveillance and safety.

We curated a diverse dataset using real leopard images from Roboflow Universe and custom toy leopard images for testing. These were carefully annotated and processed for training. Using Google Colab's GPU environment, we trained our YOLOv5 model and exported the best-performing weights. Real-time detection was then implemented using a standard webcam on a laptop, allowing us to demonstrate the system's functionality without requiring specialized hardware. A key feature of our system is its ability to send instant SMS alerts via Twilio when a leopard is detected. This makes it useful for rapid response in high-risk zones near forest boundaries. Although we faced challenges such as inconsistent data labels, training interruptions on Colab, and webcam integration issues, we overcame them through debugging, optimization, and tool integration.

Overall, this project meets its goal of creating a low-cost, deployable wildlife detection system. It showcases how open-source tools and pre-trained models can be harnessed for meaningful impact in conservation and human-wildlife conflict prevention. With further development, the system can be expanded to detect multiple species, integrate with GPS for location logging, and support real-time alerts at scale.

CRITICAL QUESTION AND ANSWERS

1. Who benefits from this?

This project benefits a range of stakeholders, especially those involved in wildlife conservation and rural safety. Forest departments and conservation agencies can use this system for early detection of leopards in buffer zones, reducing the risk of human-wildlife conflict. Rural villagers and farmers living near forests are direct beneficiaries, as timely alerts can help them avoid dangerous encounters. Additionally, researchers studying leopard behavior, migration, or population patterns can utilize the data gathered. NGOs, environmentalists, and rescue teams operating in wildlife-prone areas also gain a valuable tool for proactive monitoring.

2. What are the strengths and weaknesses?

The project's main strength lies in its use of lightweight and efficient technology like YOLOv5, which enables real-time detection with low-cost hardware like webcams. Its integration with SMS alerts through Twilio enhances responsiveness and practicality in remote areas. The use of Roboflow for dataset management also streamlines the training process. However, weaknesses include its dependency on internet access for SMS delivery and limited detection performance in low-light or occluded environments. The system is also currently focused only on leopard detection, limiting its versatility in detecting multiple species.

3. What is another alternative?

Alternative approaches to this system include the use of infrared motion sensors, which are simpler but cannot distinguish between species. Camera traps combined with manual analysis are also widely used, though they involve significant delays and human effort. More advanced alternatives like drone-based thermal imaging or deploying newer models such as YOLOv8 offer potential improvements but often at higher costs or complexity. These alternatives may provide better accuracy or broader coverage, but they may not match the affordability and simplicity of the proposed system.

4. What is the worst or best-case scenario?

In the best-case scenario, the system correctly identifies a leopard entering a village boundary and immediately sends an SMS alert. This allows forest officials or villagers to take preventive action, avoiding injuries, loss of livestock, or retaliatory killings. In the worst-case scenario, the system fails to detect a leopard due to poor camera positioning, low visibility, or model limitations. This could lead to missed alerts, delayed response, and potential harm to both humans and wildlife, undermining the purpose of early detection.

5. Where is there a need for this?

There is a pressing need for such a system in areas that lie near forests, wildlife sanctuaries, or national parks—particularly in Indian states like Maharashtra, Uttarakhand, and Karnataka, where leopard sightings and attacks are reported frequently. Agricultural regions and tribal settlements located in leopard corridors face recurring threats. Additionally, urban-fringe areas experiencing increased leopard incursions due to deforestation would greatly benefit from an automated alert system like this. Deploying such systems can fill a crucial gap in early warning infrastructure.

6. Where can we get more information?

More information on this topic can be sourced from online platforms like Roboflow's official documentation and Ultralytics' GitHub repository for YOLOv5. Research articles in journals such as *Ecological Informatics, Wildlife Society Bulletin,* and *IEEE Access* provide deeper insight into AI applications in conservation. Twilio's API documentation offers guidance on integrating SMS alerts. Additionally, government wildlife reports, newspaper archives, and conservation NGO websites often include case studies, statistics, and updates related to leopard encounters and human-wildlife conflicts.

7. What are the areas of improvement?

There are several promising areas for enhancement. One is expanding the system to detect multiple animal species, making it more useful in regions with diverse wildlife threats. Improving performance in night-time conditions using infrared or thermal cameras would make detection more reliable. Integration with GPS modules could allow location-based alerts and detection history logging. A user-friendly dashboard or mobile app could be developed for easier monitoring. These improvements would make the system more robust, adaptable, and suitable for wider deployment.

8. When would this benefit our society?

This system would greatly benefit society during times of increased leopard movement, such as after monsoon seasons or during dry months when animals venture out in search of food or water. It is also particularly useful during crop harvesting periods when humans and animals frequently share the same space. Over the long term, deploying such systems can contribute to community safety, reduce conflict-related deaths, and promote harmonious human-wildlife coexistence—especially in regions struggling with deforestation and shrinking natural habitats.

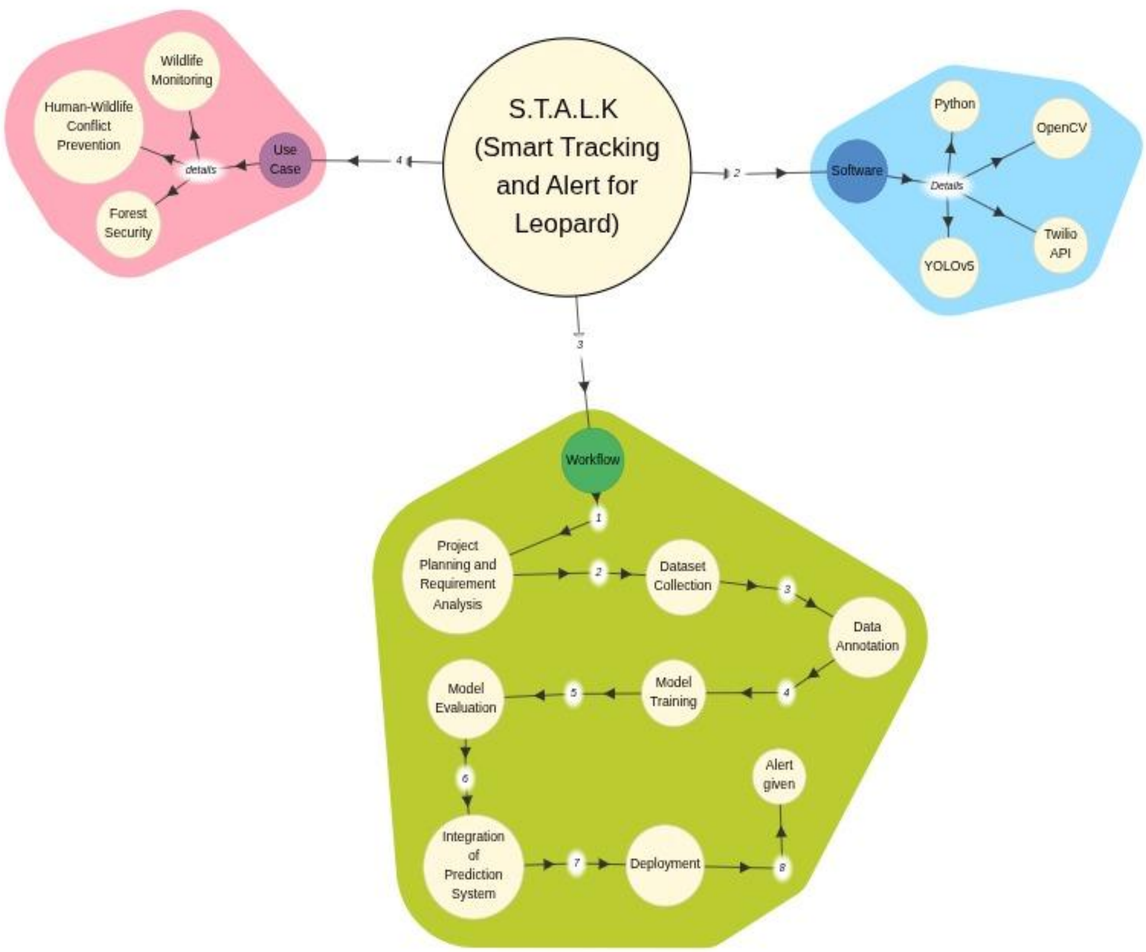
9. Why is this a problem or challenge?

Leopard intrusion into human settlements has become a growing problem due to shrinking forest cover and urban expansion into wildlife habitats. These conflicts often result in injuries, loss of livestock, and fatalities on both sides. Traditional detection methods are either delayed or lack species-specific identification. With many rural areas lacking access to real-time monitoring tools, there is an urgent need for low-cost, automated solutions. Without timely detection and alerts, the window for safe intervention is often missed, making the challenge more dangerous.

10. How does this benefit us or others?

The system provides a simple yet effective way to protect lives—both human and animal—by offering timely alerts of leopard presence. It empowers forest officials and local communities with a proactive tool for monitoring and quick response. By preventing panic and enabling organized interventions, it reduces the likelihood of retaliation against leopards, thus supporting conservation goals. On a broader level, the project demonstrates how artificial intelligence and affordable hardware can be used to address real-world challenges in safety, ecology, and rural development.

CONCEPT MAP


S.T.A.L.K
(Smart Tracking
and Alert for
Leopard)
Use Case
details
Wildlife Monitoring
Human-Wildlife Conflict Prevention
Forest Security
Software
Details
Python
OpenCV
Twilio API
YOLOv5
Workflow
Project Planning and Requirement Analysis
Dataset Collection
Data Annotation
Model Training
Model Evaluation
Integration of Prediction System
Deployment
Alert given

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